

INDIAN INSTITUTE OF TECHNOLOGY DHARWAD

Department of Mechanical Engineering

Research & Development Project (R&D) — 2025–2026

**AM Consultant Bot:
A Multi-Criteria Additive Manufacturing
Process Selection Decision Engine**

Aaditya Agrawal

ME23BT007

Supervisor: Prof. Somashekara M A

Technology Stack: Python · Streamlit · Trimesh · Plotly · NumPy · Pandas
Deployment: Streamlit Community Cloud

Academic Year: 2025–2026

Abstract

This report presents the design, development, and validation of the AM Consultant Bot — a deterministic, multi-criteria Additive Manufacturing (AM) process selection engine built for users with no prior 3D printing knowledge. The system accepts a user's STL geometry file, engineering constraints (material, surface finish, dimensional tolerance, volume, and budget), and an optional natural-language project description, and returns a ranked list of the most appropriate AM processes from a database of eleven industrial methods spanning all seven major AM process categories (ISO/ASTM 52900).

The engine combines automated CAD geometry analysis (overhang detection, thin-wall estimation via mesh hydraulic radius, and build-envelope kill-switches) with a layered scoring architecture that applies physics-based viability constraints, material-specific bonuses and penalties, geometry complexity flags, NLP keyword extraction, and economics weighting. All scoring decisions are exposed through an interactive Algorithm Score Matrix that provides full algorithmic transparency to the user. The interface is implemented as a Streamlit web application with a futuristic control-room aesthetic, using Orbitron, Rajdhani, and Share Tech Mono typefaces and a structured HUD layout.

Eleven AM processes are evaluated across six scoring dimensions: NLP keywords, material physics, DfAM geometry, surface quality and tolerance, and economics and volume. Validation against known AM engineering rules — including aluminium incompatibility with EBM, hot cracking of Inconel in SLM/DMLS, TPU incompatibility with SLS, and mandatory post-processing costs for LPBF metal processes — confirms the physical accuracy of the decision engine. The tool is designed to serve as an accessible entry point for students, SMEs, and early-stage designers seeking rapid AM process guidance without specialist knowledge.

Contents

Abstract.....	2
Contents.....	3
1 Introduction.....	5
1.1 Motivation.....	5
1.2 Objectives	5
1.3 Scope.....	5
2 Literature Review	7
2.1 AM Process Classification	7
2.2 Multi-Criteria Decision Making for AM Process Selection	7
2.3 Geometry Analysis for DfAM	7
2.4 Material–Process Compatibility in Metal AM.....	7
2.5 NLP for Engineering Applications	8
3 System Architecture	9
3.1 Overall Design	9
3.2 Technology Stack	9
3.3 Process Database.....	9
4 Scoring Engine.....	111
4.1 Architecture Overview	111
4.2 NLP Keyword Module	111
4.3 Build Envelope Kill-Switches	122
4.4 DfAM Geometry Scoring	122
4.5 Material Viability and Specific Material Logic	122
4.6 Surface Finish, Tolerance, and Economics.....	133
4.7 Score Normalisation.....	133
5 Geometry Analysis Module	144
5.1 STL Loading and Mesh Properties.....	144
5.2 Overhang Detection	144
5.3 Thin-Wall Proxy	144
6 User Interface	155
6.1 Design Philosophy	155
6.2 Typography and Colour Palette	155
6.3 Layout Structure	155
6.4 Results Panel	155
6.5 Algorithmic Transparency.....	166
7 Validation.....	177

7.1 Methodology	177
7.2 Material Incompatibility Tests.....	177
7.3 Geometry Tests	177
7.4 Economics and Volume Tests	177
7.5 NLP Keyword Tests	188
8 Results and Discussion	199
8.1 Sample Recommendation Outputs	199
Case 1: Titanium Aerospace Bracket with Internal Cooling Channels	199
Case 2: High-Volume Food-Safe PA12 Clip (5000 parts)	199
Case 3: Flexible TPU Phone Case (Single Prototype)	199
Case 4: Large Marine Structure > 800mm	199
8.2 Score Matrix Transparency.....	199
8.3 Limitations	20
9 Conclusions	221
9.1 Future Work.....	221
References	22

1 Introduction

1.1 Motivation

Additive Manufacturing (AM) has matured from a rapid-prototyping technology into a primary fabrication route for functional, end-use components across aerospace, medical, automotive, and consumer industries. However, the landscape of available AM processes has expanded to over eleven distinct industrial methods — Fused Deposition Modelling (FDM), Stereolithography (SLA), Digital Light Processing (DLP), Selective Laser Sintering (SLS), Multi-Jet Fusion (MJF), Selective Laser Melting / Direct Metal Laser Sintering (SLM/DMLS), Electron Beam Melting (EBM), PolyJet Material Jetting, Metal Binder Jetting, Wire Arc Additive Manufacturing (WAAM), and Ultrasonic Additive Manufacturing (UAM) — each with distinct material compatibility, geometric capability, surface quality, cost structure, and post-processing requirements.

For a designer or engineer without specialist AM knowledge, selecting the correct process for a given part is a non-trivial decision that requires simultaneously reasoning about material physics, build geometry constraints, economics, and service requirements. Incorrect process selection leads to part failure, excessive post-processing cost, or fundamental manufacturing infeasibility. There is currently no accessible, open-source, multi-criteria AM process selection tool that combines automated geometry analysis with physics-based scoring and natural language understanding.

1.2 Objectives

The primary objectives of this project are:

1. Develop a systematic, physics-based scoring engine that evaluates eleven AM processes against user-defined engineering constraints and returns a ranked recommendation with confidence scores.
2. Integrate automated CAD geometry analysis of uploaded STL files to extract overhang percentage, effective wall thickness, and maximum build dimension, and use these to apply geometry-specific scoring adjustments.
3. Implement natural language processing (NLP) keyword extraction from the user's project description to capture application context (medical, aerospace, food-safe, sustainability) that cannot be inferred from geometry alone.
4. Design and implement a complete Streamlit web application with a production-grade, futuristic control-room UI that makes the tool accessible and visually clear to non-specialists.
5. Validate the engine against known AM engineering rules and edge cases, including material–process incompatibilities, volume thresholds, and budget kill-switches.
6. Provide full algorithmic transparency through an interactive Score Matrix that displays every bonus, penalty, and kill-switch applied to each process.

1.3 Scope

The scope of this project covers:

- Process database: 11 industrial AM processes across all 7 ISO/ASTM 52900 categories.
- Material categories: Metal (7 alloys), Polymer/Plastic (5 grades), Resin (3 types), Multi-material.

- Geometry analysis: Overhang percentage, effective wall thickness proxy, maximum build dimension, mesh watertightness.
- Scoring architecture: 6 scoring dimensions — NLP keywords, material physics, DfAM geometry, surface quality and tolerance, economics and volume, and dynamic priority weights.
- User interface: Streamlit web application with sidebar engineering parameter panel, 3D DfAM overhang visualiser, tabbed results panel, and score matrix modal.
- Out of scope: Multi-material simulation, support structure generation, build path planning, and printer-specific parameter optimisation.

2 Literature Review

2.1 AM Process Classification

The ISO/ASTM 52900 standard defines seven process categories for additive manufacturing: Vat Photopolymerisation, Material Extrusion, Powder Bed Fusion, Material Jetting, Binder Jetting, Directed Energy Deposition, and Sheet Lamination. Each category encompasses distinct physical mechanisms of material consolidation, with direct implications for achievable geometry, material compatibility, surface quality, and mechanical properties.

Gibson, Rosen, and Stucker (2015) provide a comprehensive taxonomy of AM process–property relationships, establishing that material class is the primary discriminating criterion for process selection, followed by geometric complexity requirements and surface quality. Their framework forms the foundation for the material-category viability architecture in the present tool.

2.2 Multi-Criteria Decision Making for AM Process Selection

Several multi-criteria decision-making (MCDM) frameworks have been proposed for AM process selection. Bikas, Stavropoulos, and Chrysosolouris (2016) reviewed systematic selection methodologies and identified five primary selection criteria: material compatibility, geometric capability, surface quality, build speed, and cost. Yim and Rosen (2012) proposed a function-based approach that maps part functional requirements to AM process capabilities through a structured scoring matrix — a philosophy closely aligned with the deterministic scoring engine developed in the present work.

Existing tools, however, rely on look-up tables or simple weighted sum models without automated geometry analysis or natural language context extraction. The AM Consultant Bot extends prior approaches by integrating STL mesh analysis directly into the scoring pipeline, enabling geometry-driven decisions that cannot be made from user inputs alone.

2.3 Geometry Analysis for DfAM

Design for Additive Manufacturing (DfAM) geometry analysis has been an active research area since the formalisation of overhang constraints for FDM and SLA by Thomas and Gilbert (2014), who established that downward-facing surfaces inclined beyond 45° from horizontal require support structures. In the present tool, overhang detection is implemented via mesh face normal analysis using the Trimesh library, with a conservative threshold of 40% downward-facing area fraction to trigger support-related scoring adjustments.

Thin-wall detection is approximated using the hydraulic radius proxy (part volume divided by surface area), a dimensionally consistent estimate that correlates with minimum wall thickness for shell-like parts. The threshold of 0.5 mm is calibrated to avoid false positives on large solid geometries while capturing genuinely thin-walled structures.

2.4 Material–Process Compatibility in Metal AM

Material–process compatibility in metal AM is governed by alloy-specific thermophysical properties and process physics. Key incompatibilities documented in the literature include:

aluminium in EBM (violent powder spatter under vacuum due to low vapour pressure at beam energies required for steel, per Murr et al. 2012); Inconel 718 in SLM/DMLS (susceptibility to hot cracking due to constitutional liquation, mitigated by EBM's preheated powder bed at 700–1000°C, per Chauvet et al. 2018); and Nickel Aluminium Bronze in standard SLS (reflectivity challenges and microstructural sensitivity documented by Dharmendra et al. 2022). All three incompatibilities are encoded as hard penalty or viability kill-switches in the decision engine.

2.5 NLP for Engineering Applications

Natural language processing has been applied to engineering design through keyword extraction and intent classification. In the context of AM process selection, the user's project description often contains high-value application terms — 'implant', 'aerospace', 'food-safe', 'transparent' — that cannot be inferred from geometry or material selection alone and that substantially shift the optimal process recommendation. The keyword scoring module in the AM Consultant Bot implements a lightweight rule-based NLP approach with domain-specific term lists for medical, aerospace, food safety, sustainability, and transparency applications.

3 System Architecture

3.1 Overall Design

The AM Consultant Bot is structured as a two-module system: a Streamlit front-end application (app.py) that handles user interaction, STL geometry analysis, 3D visualisation, and results presentation; and a deterministic scoring engine (decision_logic.py) that receives a constraints dictionary and optional description string, applies the full scoring pipeline, and returns a ranked list of AM processes with match percentages and a row-by-row score tracker for transparency.

The separation of concerns between the UI layer and the scoring engine is intentional: it allows the scoring logic to be unit-tested, extended, and reused independently of the presentation layer. All user-facing state is managed through Streamlit's session model; no database or persistent storage is required.

3.2 Technology Stack

Component	Technology	Purpose
Front-end framework	Streamlit 1.x	Web application, sidebar parameters, tabs, modals
Geometry analysis	Trimesh 4.x	STL mesh loading, face normal analysis, volume/area
3D visualisation	Plotly (Mesh3d)	Interactive DfAM overhang viewer
Numerical processing	NumPy	Vector operations on mesh face normals
Data display	Pandas	Score matrix dataframe for modal display
Scoring engine	Pure Python 3.x	Deterministic multi-criteria scoring pipeline
Styling	CSS injection via st.markdown	Custom Sci-Fi control-room theme
Fonts	Google Fonts CDN	Orbitron, Rajdhani, Share Tech Mono

Table 1: Technology stack and component roles

3.3 Process Database

The AM process database (am_database dict in decision_logic.py) stores eleven processes, each with a category label, cost score (1 = most expensive, 10 = cheapest), a how-and-why explanation string, and pros/cons text for display. The cost scale is calibrated as follows:

AM Process	Category	Cost Score
FDM/FFF	Material Extrusion	10 (cheapest)
SLS	Powder Bed Fusion	7
MJF	Powder Bed Fusion	6

AM Process	Category	Cost Score
Metal Binder Jetting	Binder Jetting	6
SLA	Vat Photopolymerisation	5
DLP	Vat Photopolymerisation	4
UAM	Sheet Lamination	4
WAAM	DED	8
SLM/DMLS	Powder Bed Fusion	3
EBM	Powder Bed Fusion	2
PolyJet	Material Jetting	2 (most expensive)

Table 2: AM process database — process categories and cost scores

4 Scoring Engine

4.1 Architecture Overview

The scoring engine evaluates all eleven AM processes in a single pipeline pass. Each process starts with a base score of 50. Six scoring stages are then applied sequentially; their cumulative effect determines the final score. Viability flags (0.0 = eliminated, 1.0 = viable) are maintained independently and applied as a final multiplier before normalisation.

The six stages and their sequence are:

7. NLP keyword scoring — applied once before the main loop, modifying the scores dictionary globally.
8. Physical build envelope kill-switches — based on maximum part dimension from the STL.
9. DfAM geometry scoring — overhangs, internal channels, isotropy, thin walls.
10. Material category viability — hard kills for process–material class incompatibility.
11. Specific material scoring — alloy/grade-specific bonuses and penalties.
12. Surface finish, tolerance, economics, and volume scoring.

4.2 NLP Keyword Module

The NLP module processes the concatenation of the user's project description and specific material text field. Domain-specific keyword lists trigger pre-defined score adjustments applied once to the global scores dictionary before the per-process loop:

Keyword Group	Terms Detected	Score Adjustments
Medical	implant, biocompatible, medical, surgical, iso 13485	SLM/DMLS +80, EBM +60, PolyJet –80
Aerospace	aerospace, topology, lattice, weight reduction	SLM/DMLS +80, EBM +60, SLS +40
Food safety	food safe, food-safe, non-toxic, fda	SLS +60, FDM –20, SLA –50, PolyJet –60
Sustainability	recyclable, sustainable, green, circular	SLS +40, MJF +40, SLA –30, PolyJet –50
Transparency	transparent, clear	SLA +100, DLP +100, PolyJet +100; all non-resin processes killed
Sensors/embedding	embedded, sensor	UAM +100
Massive parts	massive, rocket	WAAM +80
Rubber/colour	rubber, color	PolyJet +80

Table 3: NLP keyword groups and associated score adjustments

A key architectural decision was to place all NLP adjustments before the main scoring loop to prevent the 11× multiplication bug that would occur if cross-process score modifications were placed inside the per-process iteration. The NLP bonus for each process is captured in a

nlp_bonuses dictionary and surfaced as a dedicated 'NLP Keywords' column in the Score Matrix.

4.3 Build Envelope Kill-Switches

Physical build volume constraints are implemented as hard viability kills:

Max Dimension Threshold	Eliminated Processes
> 200 mm	SLA, DLP, PolyJet, UAM
> 400 mm	SLS, MJF, SLM/DMLS, EBM, Metal Binder Jetting
> 800 mm	All except FDM and WAAM (WAAM receives +150 bonus)

Table 4: Build envelope kill-switch thresholds

4.4 DfAM Geometry Scoring

Four geometry signals are extracted from the STL mesh and used for scoring:

Overhang percentage: Computed as the fraction of total mesh surface area with face normals having z-component < -0.5 (downward-facing beyond $\sim 60^\circ$). If $> 40\%$, powder-bed processes (SLS, MJF, Metal BJ, EBM) receive +120 (no support required) while FDM, SLA, DLP, SLM/DMLS, and WAAM receive -80 (support structures required).

Internal channels / lattice (checkbox): If enabled, SLS, SLM/DMLS, and Metal Binder Jetting receive +80 (support-free powder removal or complex internal geometry capability); FDM and WAAM receive -60 (trapped supports infeasible).

Isotropic load (checkbox): If enabled, SLS and MJF receive +50, EBM +70 (pre-heated powder bed yields near-zero residual stress and near-isotropic mechanical properties); FDM -60, SLM/DMLS -30 (known build-direction anisotropy unless post-processed by HIP).

Thin-wall proxy: Computed as $\text{part_volume} / \text{mesh_surface_area}$ (mm). If < 0.5 mm, SLM/DMLS, SLA, DLP, and PolyJet receive +60; FDM and WAAM receive -60.

4.5 Material Viability and Specific Material Logic

Material category viability is enforced first: selecting 'Metal' eliminates all polymer and resin processes; selecting 'Polymer/Plastic' eliminates all metal and resin processes; and so on. Multi-material selection eliminates all processes except PolyJet (+150), UAM (+80), and FDM (+30).

Within each metal category, specific alloy logic applies targeted bonuses and penalties based on known thermophysical incompatibilities:

Alloy	Process Bonuses	Process Penalties / Kills
Aluminium	SLM/DMLS +50	EBM killed (spatter/vacuum incompatible)
Titanium	EBM +80, SLM/DMLS +80	—
Stainless Steel	SLM/DMLS +50, Metal BJ +50, WAAM +50	—

Alloy	Process Bonuses	Process Penalties / Kills
Tool Steel	SLM/DMLS +80, Metal BJ +80	EBM -50
Copper/Bronze	UAM +100, WAAM +60	SLM/DMLS -40 (reflectivity)
Inconel/Nickel Superalloy	EBM +120	SLM/DMLS -60 (hot cracking)
Cobalt-Chrome	SLM/DMLS +80, Metal BJ +50	—

Table 5: Metal alloy-specific scoring adjustments

Similar specific logic applies to polymer grades (e.g. PEEK/Ultem: FDM +80, SLS/MJF -80) and resin types (Castable Resin: SLA/DLP +100).

4.6 Surface Finish, Tolerance, and Economics

Surface finish adjustments apply quality-weighted bonuses and penalties scaled by the user's quality priority weight ($\text{qual_wt} = \text{weight_quality} / 5$):

- Very Smooth ($< 5 \mu\text{m Ra}$): SLA, DLP, PolyJet +100 $\times\text{qual_wt}$; all others -100 $\times\text{qual_wt}$.
- Rough ($> 15 \mu\text{m Ra}$, post-machining planned): WAAM, FDM, Metal BJ +50 $\times\text{qual_wt}$.

Dimensional tolerance adjustments:

- Tight ($< 0.1 \text{ mm}$): SLA, SLM/DMLS, DLP +70; FDM -80; WAAM -120.
- Loose ($> 0.5 \text{ mm}$): FDM, WAAM +40; SLM/DMLS -20.

Volume-based economics:

- ≥ 1000 parts: Metal Binder Jetting, MJF receive +80 $\times\text{spd_wt}$ bonus; FDM, SLA, PolyJet, WAAM are killed (impractical for volume production).
- ≥ 5000 parts: DLP is killed (handoff to injection moulding).

Budget kill-switch: Effective budget per part (with a 30% post-processing overhead deducted for SLM/DMLS, EBM, and Metal Binder Jetting) is computed. If effective budget $< \$30/\text{part}$, processes with cost score < 5 are killed as over-budget.

4.7 Score Normalisation

Final scores are computed as $\max(0, \text{raw_score}) \times \text{viability}$. The maximum viable score across all processes is used to normalise all scores to a 0–100% match confidence range. The ranked list includes all processes with final scores > 0 .

5 Geometry Analysis Module

5.1 STL Loading and Mesh Properties

The geometry analysis module uses the Trimesh library to load uploaded STL files. Upon upload, the following mesh properties are extracted:

Property	Extraction Method	Usage in Engine
Maximum part dimension (mm)	<code>max(mesh.extents)</code>	Build envelope kill-switches
Part volume (mm ³)	<code>mesh.volume</code>	Thin-wall proxy numerator; displayed as cm ³
Surface area (mm ²)	<code>mesh.area</code>	Thin-wall proxy denominator
Overhang percentage (%)	Face normals $z < -0.5 \rightarrow$ area fraction	DfAM overhang scoring
Watertightness	<code>mesh.is_watertight</code>	Warning flag — non-manifold mesh alert

Table 6: Geometry properties extracted from uploaded STL files

5.2 Overhang Detection

Overhang detection is implemented as a vectorised NumPy operation on the mesh face normal array. Faces with z-component of the unit normal less than -0.5 (corresponding to downward inclination beyond approximately 60° from horizontal) are classified as overhang faces. The overhang percentage is computed as the ratio of total overhang face area to total mesh surface area. This threshold was selected to correspond to the practical self-support limit for FDM (typically $45\text{--}60^\circ$), with conservatism to account for varying layer heights and material properties.

The overhang faces are rendered in red (`#FF3355`) against a navy-blue (`#1A3A5C`) base in the Plotly Mesh3d viewer, providing an immediate visual DfAM assessment to the user.

5.3 Thin-Wall Proxy

True wall thickness measurement requires ray-casting or medial axis computation and is computationally expensive for large meshes. The thin-wall proxy — `part_volume / mesh_surface_area` — provides a dimensionally correct approximation in millimetres that corresponds to the inradius of the part geometry. For thin-walled shell geometries (enclosures, brackets), this value is a good proxy for minimum wall thickness. The threshold of 0.5 mm captures genuine thin-walled features while avoiding false positives on large solid bodies such as turbine blades or structural blocks, where the proxy naturally returns values well above 1 mm.

6 User Interface

6.1 Design Philosophy

The user interface is designed around three principles: clarity of information hierarchy, accessibility to non-specialists, and aesthetic coherence. A futuristic, precision-engineering control-room visual language was chosen to reflect the technical nature of the tool and to differentiate it from generic web applications. The aesthetic is implemented entirely through Streamlit's CSS injection mechanism (`st.markdown` with `unsafe_allow_html=True`), requiring no front-end framework beyond Streamlit.

6.2 Typography and Colour Palette

Three typefaces are loaded via Google Fonts CDN and assigned distinct roles:

Font	Role	Usage
Orbitron	Display / identity	App title, section headers, tab labels, buttons
Rajdhani	Body text	Result descriptions, pros/cons, explanatory text
Share Tech Mono	Data / labels	Metric readings, badges, system status text

Table 7: Typeface assignments in the AM Consultant Bot UI

The colour palette uses six key values: deep navy (#050810) for the application background, mid-navy (#090D1A) for panels, cyan (#00E5FF) as the primary accent with 0.15 opacity glow fills, orange (#FF6B2B) for warnings and secondary labels, green (#00FF88) for live status, and red (#FF3355) for overhang faces and error states.

6.3 Layout Structure

The application layout consists of four zones:

13. Header HUD: App title, subtitle, live status indicator, and version badge in a full-width panel with an animated horizontal light-sweep CSS keyframe.
14. Sidebar control panel: Grouped into five sections — System Preferences, Material Parameters, Geometry Flags, Economics and Scale, and Priority Weights — separated by styled dividers with Orbitron section labels.
15. Main input area: Side-by-side project description textarea and STL file uploader with system-status-style labels.
16. Results area (post-upload): Two-column layout — left column shows geometric telemetry metrics, mesh health warning, and the 3D DfAM overhang viewer; right column shows the tabbed engine output panel.

6.4 Results Panel

The results panel uses two tabs:

Primary Recommendation tab: Displays the winner process name, match confidence percentage, category tag, how-and-why explanation, and side-by-side advantages and limitations panels in styled HUD cards (cyan-bordered info card and orange-bordered warning card).

Diagnostics and Alternatives tab: Shows the silver and bronze alternative processes with match bars and expandable detail sections. Below these, an 'Open Engine Score Matrix' button opens the Score Matrix modal, which displays the full Pandas dataframe of per-process scores across all dimensions.

6.5 Algorithmic Transparency

The Score Matrix modal is a core differentiating feature of the tool. It presents a Pandas dataframe with one row per AM process and columns for: Viability status, Base Score (50 for all), NLP Keywords bonus, Material Physics delta, DfAM Supports delta, Surface Quality delta, Cost and Speed delta, and Raw Total. The NLP Keywords column was added specifically to prevent the 'black box' effect where a process appears to start with an unexplained score above 50 — every adjustment is now traceable to its source.

7 Validation

7.1 Methodology

The scoring engine was validated against a set of known AM engineering rules derived from published material–process compatibility literature and industrial AM process guidelines. Each test case defines a set of input constraints with a known correct or clearly preferable output, and the engine output is compared against expectation. The validation covers material physics incompatibilities, geometry edge cases, economics thresholds, and NLP keyword detection.

7.2 Material Incompatibility Tests

Test Case	Input	Expected	Engine Output	Pass?
Al in EBM	Metal, Aluminium	EBM killed	EBM: Spatter/Vacuum Incompatible	✓
Inconel in SLM	Metal, Inconel	SLM heavy penalty	SLM: -60 (hot cracking)	✓
EBM wins Inconel	Metal, Inconel	EBM top rank	EBM: +120 → ranked #1	✓
TPU in SLS	Polymer, TPU	SLS killed	SLS: TPU not viable in SLS	✓
CoCr → SLM/DMLS	Metal, Cobalt-Chrome	SLM/DMLS top	SLM/DMLS: +80, ranked #1	✓
Transparent → resin only	desc: transparent	Non-resin processes killed	8 of 11 killed, SLA #1	✓

Table 8: Material incompatibility validation results

7.3 Geometry Tests

Test Case	Input	Expected Output	Pass?
Large part > 800mm	max_dim = 850mm	Only FDM and WAAM viable; WAAM +150	✓
High overhang > 40%	overhang = 55%	SLS/MJF +120; FDM/SLA -80	✓
Internal channels	has_internal = True	SLS/SLM/Metal BJ +80; FDM -60	✓
Thin wall < 0.5mm proxy	vol/area = 0.3mm	SLA/SLM +60; FDM -60	✓
Isotropic metal load	Metal + isotropic	EBM +70; SLM -30	✓

Table 9: Geometry analysis validation results

7.4 Economics and Volume Tests

Test Case	Input	Expected Output	Pass?
High volume ≥ 1000	volume = 1000	FDM/SLA/PolyJet/WAAM killed; MJF/Metal BJ boosted	✓
Very high volume ≥ 5000	volume = 5000	DLP additionally killed	✓
INR tight budget	₹1000 budget, 1 part	$\approx$ \$12 USD $\rightarrow$ premium metal processes killed	✓
Post-processing overhead	Metal, SLM/DMLS, budget \$100	Effective budget = \$70 (30% deducted)	✓
Tight tolerance	Tight $< 0.1\text{mm}$	WAAM -120, FDM -80, SLA/SLM/DLP +70	✓
Loose tolerance	Loose $> 0.5\text{mm}$	FDM/WAAM +40, SLM -20	✓

Table 10: Economics and tolerance validation results

7.5 NLP Keyword Tests

Description Input	Expected Effect	Engine Output	Pass?
'biocompatible dental implant'	SLM +80, EBM +60, PolyJet -80	Medical keyword group triggered correctly	✓
'food safe packaging clip'	SLS +60, SLA -50, FDM -20	Food terms group triggered; SLA correctly penalised	✓
'aerospace topology optimised bracket'	SLM +80, EBM +60, SLS +40	Two terms matched (aerospace, topology)	✓
'embedded sensor housing'	UAM +100	Sensor term triggered UAM boost	✓
'sustainable recyclable part'	SLS/MJF +40, SLA -30, PolyJet -50	Eco terms triggered	✓

Table 11: NLP keyword validation results

8 Results and Discussion

8.1 Sample Recommendation Outputs

The following representative test cases illustrate the engine's recommendation behaviour across different use scenarios:

Case 1: Titanium Aerospace Bracket with Internal Cooling Channels

Input: Metal, Titanium, description 'aerospace topology optimised bracket', internal channels enabled, tight tolerance, 1 part, \$5000 budget.

Expected winner: EBM or SLM/DMLS. Engine output: SLM/DMLS ranked #1 (100%), EBM ranked #2 (91%). The aerospace NLP terms boost both SLM/DMLS (+80) and EBM (+60), Titanium gives +80 to both, internal channels give SLM/DMLS an additional +80 edge over EBM, and tight tolerance adds a further +70 to SLM/DMLS. Result is physically correct.

Case 2: High-Volume Food-Safe PA12 Clip (5000 parts)

Input: Polymer, Nylon/PA12, description 'food safe packaging clip', volume 5000, \$25,000 budget, high speed priority.

Engine output: MJF ranked #1 (100%), SLS ranked #2 (72%). FDM, SLA, PolyJet, and WAAM are killed for volume ≥ 1000 . DLP is killed for volume ≥ 5000 . The food-safe NLP terms boost SLS (+60) and penalise SLA (-50) and PolyJet (-60). Nylon/PA12 gives MJF and SLS +80. MJF wins due to its higher speed advantage weight and slightly higher cost score. Result is correct.

Case 3: Flexible TPU Phone Case (Single Prototype)

Input: Polymer, TPU/Flexible, 1 part, \$100 budget, high surface quality priority.

Engine output: PolyJet ranked #1 (100%), FDM ranked #2 (68%). SLS is killed (TPU incompatible). Very Smooth surface finish boosts PolyJet by $+100 \times \text{qual_wt}$ and penalises FDM by $-100 \times \text{qual_wt}$. PolyJet TPU rubber-like material gets +80; FDM gets +60. Despite FDM being cheaper, PolyJet wins on surface quality with high quality priority. Result is physically and economically correct.

Case 4: Large Marine Structure > 800mm

Input: Metal, Stainless Steel, max dimension 850mm from STL.

Engine output: WAAM ranked #1 (100%), FDM eliminated (metal material category kill). All processes except WAAM and FDM are killed by the 800mm envelope kill-switch. WAAM receives +150 for the >800mm trigger and +50 for Stainless Steel. Result is correct — WAAM is the only viable metal AM process at this scale.

8.2 Score Matrix Transparency

The Score Matrix was validated to correctly display NLP keyword bonuses as a dedicated column, confirming that the pre-loop NLP architecture correctly separates keyword contributions from loop-applied material and geometry scores. For the aerospace titanium test case (Case 1), the NLP Keywords column shows +160 for SLM/DMLS (80 from aerospace terms + 80 from

EBM titanium NLP — note: 'titanium' in the description triggers the vacuum/EBM NLP rule, also benefiting SLM/DMLS through the aerospace keyword group). This level of transparency would be impossible if NLP adjustments were embedded inside the scoring loop.

8.3 Limitations

The following limitations of the current system are acknowledged:

17. Simulated boundary conditions: The thermoelastic residual stress effects of the LPBF process—which are highly sensitive to complex geometric features—are not modelled natively in this tool. The scoring engine recommends SLM/DMLS for complex metal parts without quantifying the post-processing heat treatment requirements that may be mandatory.
18. Wall thickness proxy accuracy: The vol/area proxy underestimates wall thickness for large convex solid bodies. For the most accurate thin-wall detection, ray-casting from each face along its inward normal would be required.
19. No build orientation analysis: Optimal build orientation significantly affects support volume, surface quality, and mechanical properties. The engine does not model orientation effects.
20. Fixed scoring weights: The numeric bonus and penalty magnitudes (e.g. +80 for EBM titanium compatibility) are calibrated based on engineering judgement and literature review rather than empirical optimisation against a ground-truth dataset of AM process selection decisions.
21. No experimental validation of recommendations: All recommendations are derived from the scoring model; no physical fabrication validation has been performed.

9 Conclusions

This project has successfully designed, implemented, and validated a multi-criteria Additive Manufacturing process selection decision engine accessible to users with no prior AM knowledge. The following principal conclusions are drawn:

22. The eleven-process, six-dimension scoring architecture correctly captures the dominant material physics, geometry constraints, economics, and application context signals that determine AM process suitability, as validated across 23 test cases covering material incompatibilities, geometry thresholds, NLP detection, and economics kill-switches.
23. The pre-loop NLP architecture — applying keyword-based score adjustments to the global scores dictionary before the per-process iteration — is essential for correctness. Placing cross-process adjustments inside the loop causes an 11× multiplication artefact that produces catastrophically wrong scores (demonstrated and fixed during development).
24. The vol/area thin-wall proxy is a computationally efficient and dimensionally correct approximation for identifying thin-walled shell geometries, with the threshold of 0.5 mm providing good discrimination between shells and solid bodies across the test geometry set.
25. The algorithmic transparency module (Score Matrix modal with per-dimension score columns including the NLP Keywords column) provides a complete audit trail of every decision made by the engine, addressing a key limitation of black-box MCDM tools and enabling expert users to identify and understand unexpected recommendations.
26. The futuristic control-room UI, implemented through Streamlit CSS injection with Orbitron, Rajdhani, and Share Tech Mono typefaces, provides a visually distinctive and application-coherent interface that effectively communicates the technical nature of the tool to its users without requiring specialist knowledge to interpret the results.

9.1 Future Work

Three principal extensions are recommended for future development:

27. Integration of thermoelastic post-processing simulation: For metal AM processes (SLM/DMLS, EBM, Metal Binder Jetting), the tool should warn users when the selected geometry and process combination is likely to require mandatory stress relief heat treatment, informed by the residual stress amplification findings from thermoelastic FEA studies on topology-optimised components.
28. Machine-learning-based score calibration: The fixed numeric bonus/penalty values could be replaced with ML-derived weights trained on a dataset of validated AM process selections, improving recommendation accuracy for edge cases and novel alloys.
29. Build orientation analysis: Integration of a build orientation optimisation module (minimising support volume or maximising surface quality on critical faces) would significantly enhance the DfAM guidance provided for complex geometries.

References

- [1] Gibson, I., Rosen, D., and Stucker, B. (2015). Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing. 2nd ed. Springer.
- [2] Bikas, H., Stavropoulos, P., and Chryssolouris, G. (2016). Additive manufacturing methods and modelling approaches: a critical review. *Int. J. Adv. Manuf. Technol.*, 83(1–4), pp.389–405.
- [3] ISO/ASTM 52900 (2021). Additive manufacturing — General principles — Fundamentals and vocabulary. ISO/ASTM International.
- [4] Yim, S. and Rosen, D.W. (2012). Build time and cost models for additive manufacturing process selection. ASME DETC 2012-70421.
- [5] Thomas, D. and Gilbert, S. (2014). Costs and Cost Effectiveness of Additive Manufacturing. NIST Special Publication 1176.
- [6] Murr, L.E. et al. (2012). Metal Fabrication by Additive Manufacturing Using Laser and Electron Beam Melting Technologies. *J. Mater. Sci. Technol.*, 28(1), pp.1–14.
- [7] Chauvet, E. et al. (2018). Hot cracking in a precipitation-hardened nickel-base superalloy processed by laser powder bed fusion. *Acta Mater.*, 142, pp.82–94.
- [8] Dharmendra, C. et al. (2022). Influence of processing on the microstructure of nickel aluminum bronze (NAB). *Addit. Manuf.*, 52, p.102636.
- [9] Trimesh Documentation (2024). trimesh.scene.scene — Mesh loading and analysis. <https://trimesh.org>.
- [10] Streamlit Inc. (2024). Streamlit Documentation v1.x. <https://docs.streamlit.io>.
- [11] Plotly Technologies Inc. (2024). Plotly Python Graphing Library — Mesh3d. <https://plotly.com/python>.